

More Novel Data Streams

Mobility, weather, and news, borrowing what the field already uses

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A different question

The streams so far (search, Wikipedia, wastewater) mostly answer “**how much disease is here now?**” These three answer different questions:

- **Mobility:** where is it *coming from*?
- **Weather:** what is *driving* it?
- **News:** what is *emerging* that has no data yet?

Honest framing

These are streams other groups pioneered: **GLEAM** (Vespignani’s lab) for mobility, **Shaman et al.** for weather, **HealthMap / ProMED** for news. We credit them, and we show how to reach *free, scrapable* versions with an agent.

Human mobility

Weather

News alerts

Pulling it together

Human mobility

Disease travels with people, not with distance

Epidemics spread along how people actually move. Big, well-connected places seed smaller ones (the gravity / metapopulation intuition behind flu spread).

- **Ground** movement (commuting) drives local, radial spread between nearby counties.
- **Air** travel drives long-range importation, seeding distant places.

So two questions: which neighbors will seed me next, and which far-off places import risk to me?

The gold standard: GLEAM (Vespignani / MOBS lab)

The most established mobility-driven epidemic model, and worth knowing by name.

- **Air travel:** the **OAG** and **IATA** databases, roughly 3,800 airports and 4M+ connections.
- **Ground:** a **commuting** database from 40+ national statistics offices, 78,000+ regions, 5M+ connections, mapped with Voronoi cells around airports.
- **Population:** WorldPop / Gridded Population of the World.

The catch for a hands-on class

OAG, IATA, and SafeGraph are **licensed**. We cannot scrape them live. So we use **free proxies** and keep cached snapshots of the licensed ones.

Ground mobility you can actually scrape

- **US Census LODES** (origin-destination commuting) is free and detailed: who lives where and works where, down to the census tract.
- **Google COVID-19 Community Mobility** (archived 2020–2022) and the **ODT Flow Explorer** are free downloads; **SafeGraph** is licensed, so cached.

Prompt

```
Using US Census LODES origin-destination data, for Fulton
County, GA, rank the counties that send the most commuters
in. Return the top 10 as a tidy table. These are the
neighbors whose outbreaks tend to reach us first.
```

Air mobility you can actually scrape

- **OpenSky Network** exposes real flight data (ADS-B) through a free API, a workable stand-in for OAG/IATA to rank inbound connections.
- Cross-check which countries a place **exchanges** with using Wikipedia (a trick from our own planning), then reason about importation risk.

Prompt

Using the OpenSky Network API, list arrivals into Atlanta (ATL) over the past month and tally them by origin airport and country. Rank the countries that connect to ATL the most, as a proxy for air-importation risk.

Weather

Weather as a modulator, not a case count

Weather does not count cases; it **shifts the conditions** for transmission.

- **Influenza: absolute humidity** is the dominant driver. Shaman et al. showed virus survival and transmission fall as absolute humidity rises, and built humidity-driven flu forecasts.
- **Vector-borne (dengue, Zika): temperature and rainfall** govern the mosquito life cycle, so they modulate outbreak timing and size.

Mauricio picks up the modeling of weather-as-modulator this afternoon; here we just get the data.

Weather you can actually scrape

- **Open-Meteo** gives free historical and forecast weather with **no key**: ideal for an agent.
- **ERA5** (Copernicus) is the research-grade reanalysis (temperature and dew point at 0.25 degrees) behind most humidity studies.

Prompt

Using the Open-Meteo historical API (no key), pull daily temperature and dew point for Atlanta since 2016. Compute weekly absolute humidity from them, and overlay it on flu ED visits so I can see whether low humidity precedes flu season.

News alerts

News: catching what has no data yet

For a genuinely **emerging** outbreak, there is no search history and no wastewater site. The earliest signal is a report.

- **ProMED** (human-curated, since 1994) and **HealthMap** (news wires, RSS, ProMED, WHO, auto-parsed and geocoded) are the established event-based systems. ProMED flagged COVID roughly a day before official notification.
- These feed WHO's EIOS and are used by CDC and WHO.

*This is also a stream in Mauricio and Kogan's multi-source COVID forecasting:
searches + news alerts + GLEAM.*

GDELT monitors world news in 100+ languages, is 100% free, and has an API, which makes it perfect for an agent. Or point the agent at a news filter directly.

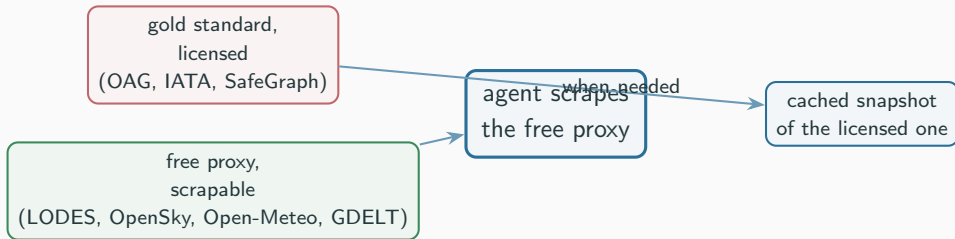
Prompt

```
Using the GDELT DOC 2.0 API, find news articles from the  
past month that mention dengue outbreaks. Return date,  
country, and headline; then summarize which countries are  
newly reporting activity, as an emerging-outbreak watch.
```

Noisy by nature: news reflects attention and reporting, so a human read is part of the loop.

Pulling it together

The pattern behind all three



- Same five-step loop, same verify habit as every other stream.
- These are **context** signals (spread, drivers, emergence), noisier than a local case proxy. Treat them as complements, and validate.

- **This afternoon:** the network / ARGONet idea borrows signal from neighbors, which is exactly what **ground mobility** tells you to do. Mauricio also covers **weather** and **news** for emerging outbreaks.
- **Day 3 capstone:** pick the streams that fit your question. Importation risk? Use air mobility. Seasonality? Weather. Emerging threat? News.

You now know the field's established streams by name, and a free, agent-scrapable way into each.

- **Mobility / GLEAM:** Balcan et al., *Multiscale mobility networks and the spatial spreading of infectious diseases*, PNAS 2009; the GLEAM Project (MOBS lab, Northeastern; Vespignani). Data: OAG, IATA, national commuting statistics, WorldPop.
- **Free mobility proxies:** US Census LODES; OpenSky Network; Google COVID-19 Community Mobility Reports (archived); ODT Flow Explorer.
- **Weather:** Shaman & Kohn (PNAS 2009) and Shaman et al. (PLoS Biology 2010; PNAS 2012) on absolute humidity and influenza. Data: Open-Meteo; ERA5 (Copernicus).
- **News:** ProMED (ISID); HealthMap (Brownstein / Freifeld); WHO EIOS; GDELT. Multi-source forecasting: Kogan, Santillana et al., *Science Advances* 2021.

Questions?

Repo: `github.com/Shihao-Yang/sismid2026`